MTE544 Fall 2026
Assignment 1

1 Mobile Manipulator [25 pts]

Solution to (a)

The transformation matrices between subsequent frames are given below. The matrix G_b^a maps coordinates from frame $\{b\}$ to frame $\{a\}$.

$$G_E^2 = \begin{bmatrix} 0 & 1 & l_2 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad G_2^1 = \begin{bmatrix} \cos\gamma & -\sin\gamma & l_1 \\ \sin\gamma & \cos\gamma & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$G_1^B = \begin{bmatrix} \cos\alpha & -\sin\alpha & l_B \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad G_B^s = \begin{bmatrix} \cos\theta & -\sin\theta & x \\ \sin\theta & \cos\theta & y \\ 0 & 0 & 1 \end{bmatrix}$$

The frame $\{E\}$ is rotated by $\psi = -\pi/2$ relative to $\{2\}$, which gives the rotation block of G_E^2. The overall transformation from $\{E\}$ to $\{s\}$ is

$$G_E^s = G_B^s\, G_1^B\, G_2^1\, G_E^2.$$

This product was simplified in MATLAB with `simplify`. With $\phi = \theta + \alpha + \gamma$, the result is

$$G_E^s = \begin{bmatrix} \sin\phi & \cos\phi & x + l_B\cos\theta + l_1\cos(\theta+\alpha) + l_2\cos\phi \\ -\cos\phi & \sin\phi & y + l_B\sin\theta + l_1\sin(\theta+\alpha) + l_2\sin\phi \\ 0 & 0 & 1 \end{bmatrix}.$$

Solution to (b)

The points were computed in MATLAB by substituting the given values into G_E^s and multiplying by each point in homogeneous coordinates, $p^s = G_E^s \begin{bmatrix} p^E \\ 1 \end{bmatrix}$. The results are

$$p_1^s = \begin{bmatrix} 2.2659 \\ -0.8248 \end{bmatrix}, \qquad p_2^s = \begin{bmatrix} 3.1024 \\ -1.5666 \end{bmatrix}.$$

Solution to (c)

The point q^B is given in $\{B\}$. The transformation from $\{E\}$ to $\{B\}$ is

$$G_E^B = G_1^B\, G_2^1\, G_E^2.$$

The point is expressed in $\{E\}$ by inverting this transformation. This was evaluated in MATLAB:

$$\begin{bmatrix} q^E \\ 1 \end{bmatrix} = \left(G_E^B\right)^{-1} \begin{bmatrix} q^B \\ 1 \end{bmatrix}$$

The result is

$$q^E = \begin{bmatrix} -0.7418 \\ 0.3542 \end{bmatrix}.$$

2 Differential-Drive Simulation [25 pts]

Model

The robot is modelled with the unicycle kinematics $\dot{x} = v\cos\theta$, $\dot{y} = v\sin\theta$, $\dot{\theta} = \omega$, which were integrated with `odeint` from the initial pose $[0,\ 0,\ 0]^\top$ over 30 s with an output timestep of 0.1 s.

Solution to (a)

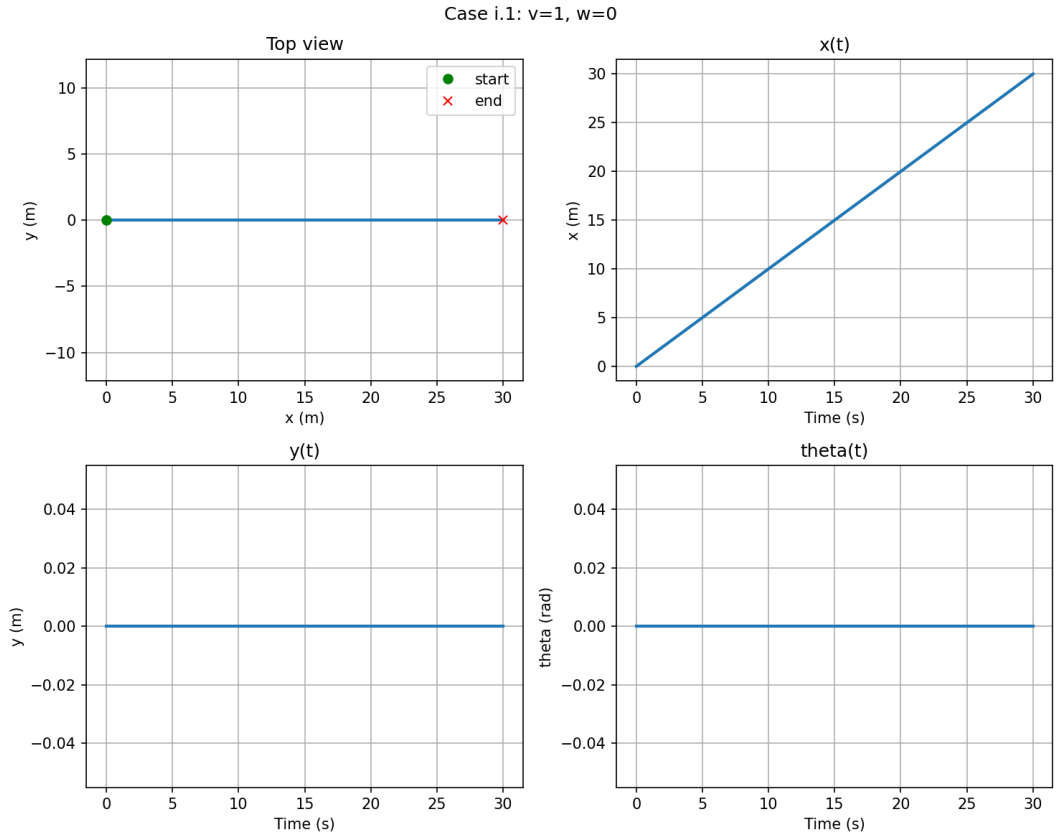


Figure 1: Case i.1: $v = 1$ m/s, $\omega = 0$ rad/s. Top view of the trajectory and x, y, θ against time.

Case i.1: The robot drives a straight line along X_S. The position x grows linearly to 30 m, while y and θ stay at zero.

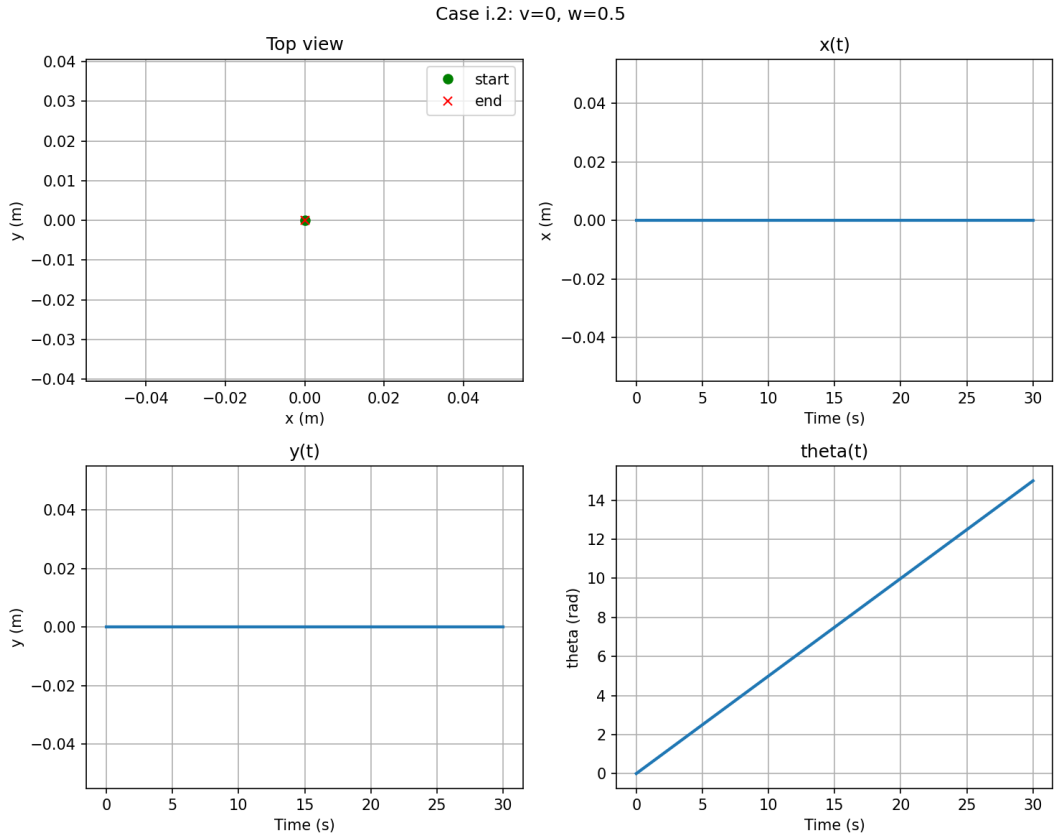


Figure 2: Case i.2: $v = 0$ m/s, $\omega = 0.5$ rad/s. Top view of the trajectory and x, y, θ against time.

Case i.2: The robot spins in place at the origin. Only θ changes, growing linearly to 15 rad, which is about 2.4 turns.

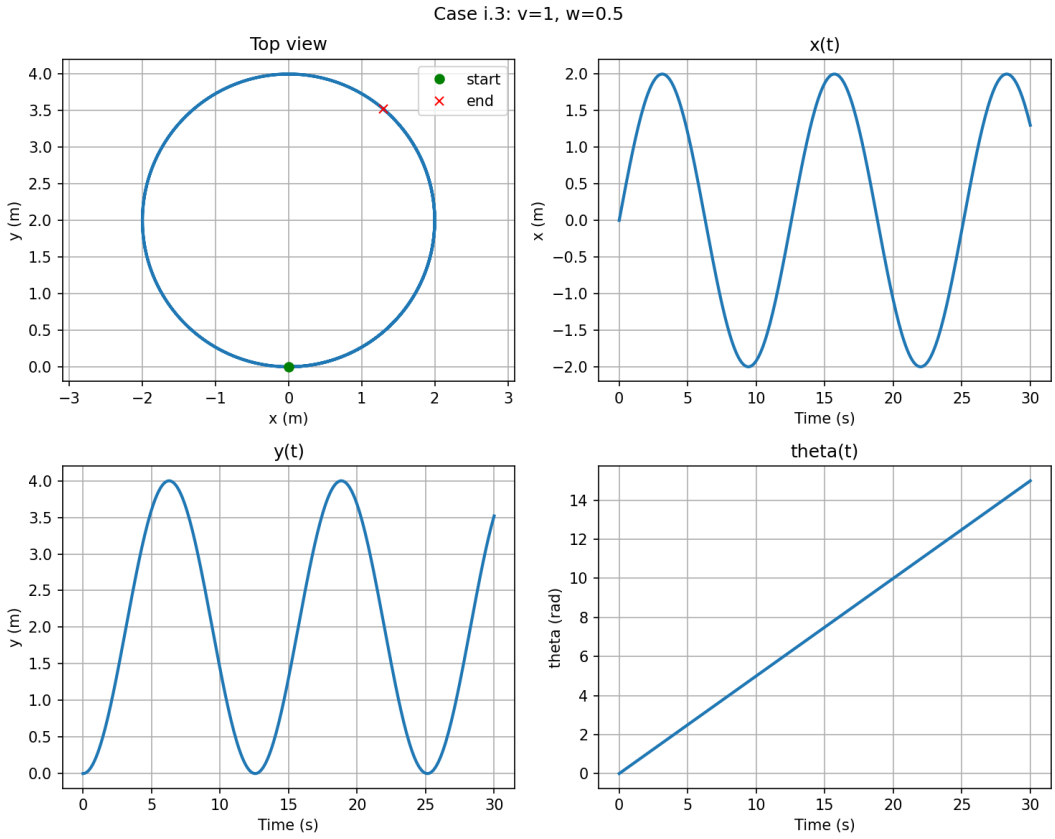


Figure 3: Case i.3: $v = 1$ m/s, $\omega = 0.5$ rad/s. Top view of the trajectory and x, y, θ against time.

Case i.3: The robot drives a circle of radius $v/\omega = 2$ m, centered at $(0,\ 2)$. One lap takes $2\pi/\omega \approx 12.6$ s, so the 30 s run covers about 2.4 laps. The signals x and y are sinusoids and θ grows linearly.

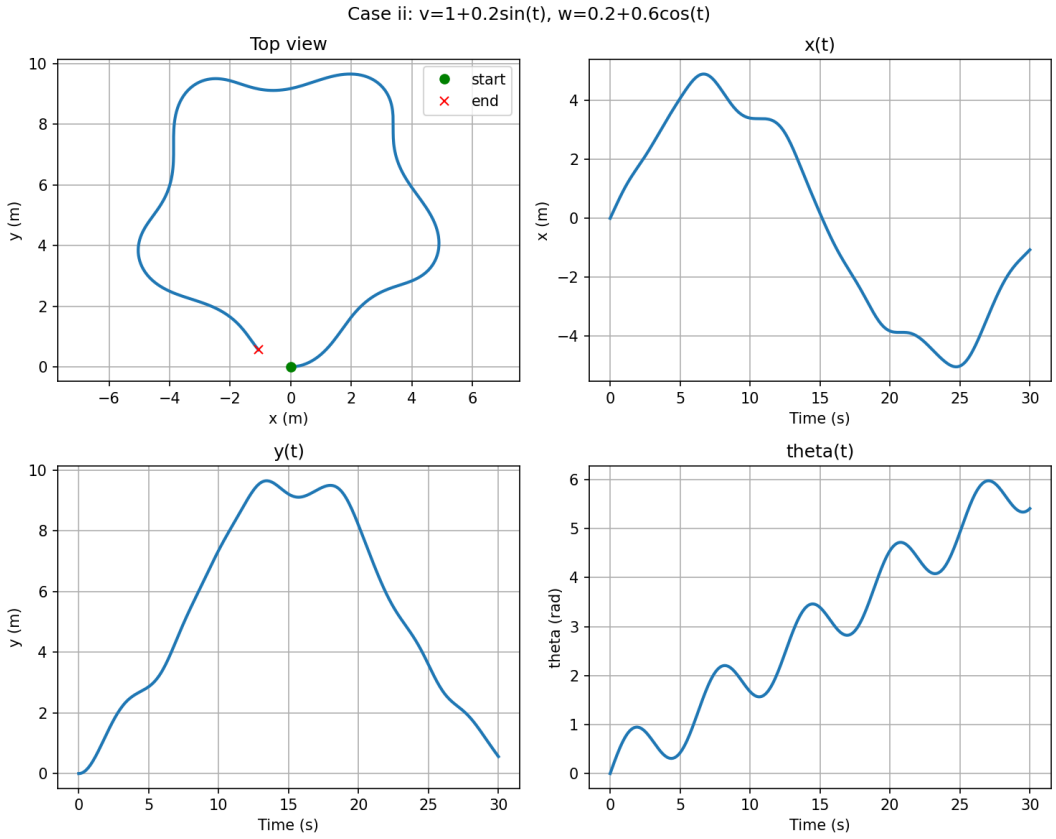


Figure 4: Case ii: $v = 1 + 0.2\sin(t)$ m/s, $\omega = 0.2 + 0.6\cos(t)$ rad/s. Top view of the trajectory and x, y, θ against time.

Case ii: Integrating $\omega(t)$ gives $\theta(t) = 0.2t + 0.6\sin(t)$. The heading therefore drifts upward at 0.2 rad/s with an oscillation on top. This produces a wavy loop about 10 m across that nearly returns to the start after 30 s. The oscillating $v(t)$ and $\omega(t)$ cause the wiggles in the path.

Solution to (b)

The wheel ground speeds are ru_r and ru_l. The robot speed is their average, and the angular velocity is their difference divided by the track:

$$v = \frac{r(u_r + u_l)}{2}, \qquad \omega = \frac{r(u_r - u_l)}{T}$$

Inverting gives

$$u_r = \frac{1}{r}\left(v + \frac{T\omega}{2}\right), \qquad u_l = \frac{1}{r}\left(v - \frac{T\omega}{2}\right)$$

With $T = 0.3$ m and $r = 0.15$ m:

Case	$[v,\ \omega]$	u_l [rad/s]	u_r [rad/s]
i.1	$[1,\ 0]$	6.67	6.67
i.2	$[0,\ 0.5]$	-0.50	0.50
i.3	$[1,\ 0.5]$	6.17	7.17

In case i.1 both wheels spin at the same speed. In case i.2 the wheels spin in opposite directions, so the robot rotates in place.

If the inputs were wheel speeds, only the start of the dynamics function would change. The function would take $[u_l,\ u_r]$ as input and first convert it to $v = r(u_r + u_l)/2$ and $\omega = r(u_r - u_l)/T$. The state equations, the integration, and the plots would stay the same.

3 3-Omniwheel Simulation [50 pts]

Solution to (a)

Single wheel

$$v^w = \begin{bmatrix} v_x^w \\ v_y^w \end{bmatrix} = \begin{bmatrix} v_{\text{drive}} \\ v_{\text{slide}} \end{bmatrix}$$

$$u = \frac{v_{\text{drive}}}{r} = \frac{v_x^w}{r}$$

Relate

$$\dot{q} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix}$$

with $(v_x^i, v_y^i) =: (v^i)^{w_i}$.

Coordinate TF for velocity

$$(v^i)^s = \hat{\omega} R_r^s p^r + v \tag{1}$$

$$(v^i)^s = R_{w_i}^s (v_i)^{w_i} = R_r^s R_{w_i}^r v_i^{w_i} \tag{2}$$

where

$$R_r^s = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, \qquad R_{w_i}^r = \begin{bmatrix} \cos\beta_i & -\sin\beta_i \\ \sin\beta_i & \cos\beta_i \end{bmatrix}$$

Combine (1) and (2):

$$v_i^{w_i} = (R_{w_i}^r)^\top (R_r^s)^\top \left(\hat{\omega} R_r^s p^r + v\right)$$

$$v_i^{w_i} = \begin{bmatrix} \omega(x_i \sin\beta_i - y_i \cos\beta_i) + \dot{x}\cos(\theta+\beta_i) + \dot{y}\sin(\theta+\beta_i) \\ \omega(x_i \cos\beta_i + y_i \sin\beta_i) - \dot{x}\sin(\theta+\beta_i) + \dot{y}\cos(\theta+\beta_i) \end{bmatrix}$$

$$u_i = \frac{1}{r} v_{\text{drive}} = \frac{1}{r} v_x^i = \frac{1}{r} g_i(\theta)\dot{q}$$

where

$$g_i = \begin{bmatrix} \cos(\theta+\beta_i) & \sin(\theta+\beta_i) & x_i \sin\beta_i - y_i \cos\beta_i \end{bmatrix}$$

Wheel positions in $\{r\}$ and angles

$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} l \\ 0 \end{bmatrix}, \qquad \beta_1 = \frac{\pi}{2}$$

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} -\frac{l}{2} \\ \frac{\sqrt{3}}{2} l \end{bmatrix}, \qquad \beta_2 = \frac{\pi}{2} + \frac{2\pi}{3} = \frac{7\pi}{6}$$

$$\begin{bmatrix} x_3 \\ y_3 \end{bmatrix} = \begin{bmatrix} -\frac{l}{2} \\ -\frac{\sqrt{3}}{2} l \end{bmatrix}, \qquad \beta_3 = \frac{\pi}{2} + \frac{4\pi}{3} = \frac{11\pi}{6}$$

Simplification of g_i

$$x_1 \sin\beta_1 - y_1 \cos\beta_1 = l, \quad x_2 \sin\beta_2 - y_2 \cos\beta_2 = l, \quad x_3 \sin\beta_3 - y_3 \cos\beta_3 = l$$

The third entry is l for every wheel. Therefore,

$$g_i = \begin{bmatrix} \cos(\theta+\beta_i) & \sin(\theta+\beta_i) & l \end{bmatrix}$$

Since $\omega = \dot{\theta}$, every wheel contributes the same term $l\dot{\theta}$ to u_i. The wheels are mounted tangentially, so each one sees the rotation equally. Stacking $u_i = \frac{1}{r} g_i(\theta)\dot{q}$ for the three wheels, with the β_i values above, gives

$$u = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = G(\theta)\dot{q} = \frac{1}{r} \begin{bmatrix} \cos\left(\theta + \frac{\pi}{2}\right) & \sin\left(\theta + \frac{\pi}{2}\right) & l \\ \cos\left(\theta + \frac{7\pi}{6}\right) & \sin\left(\theta + \frac{7\pi}{6}\right) & l \\ \cos\left(\theta + \frac{11\pi}{6}\right) & \sin\left(\theta + \frac{11\pi}{6}\right) & l \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix}$$

Solution to (b) I

The control input is

$$u = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} \text{ rad/s.}$$

The resulting robot motion is shown below.

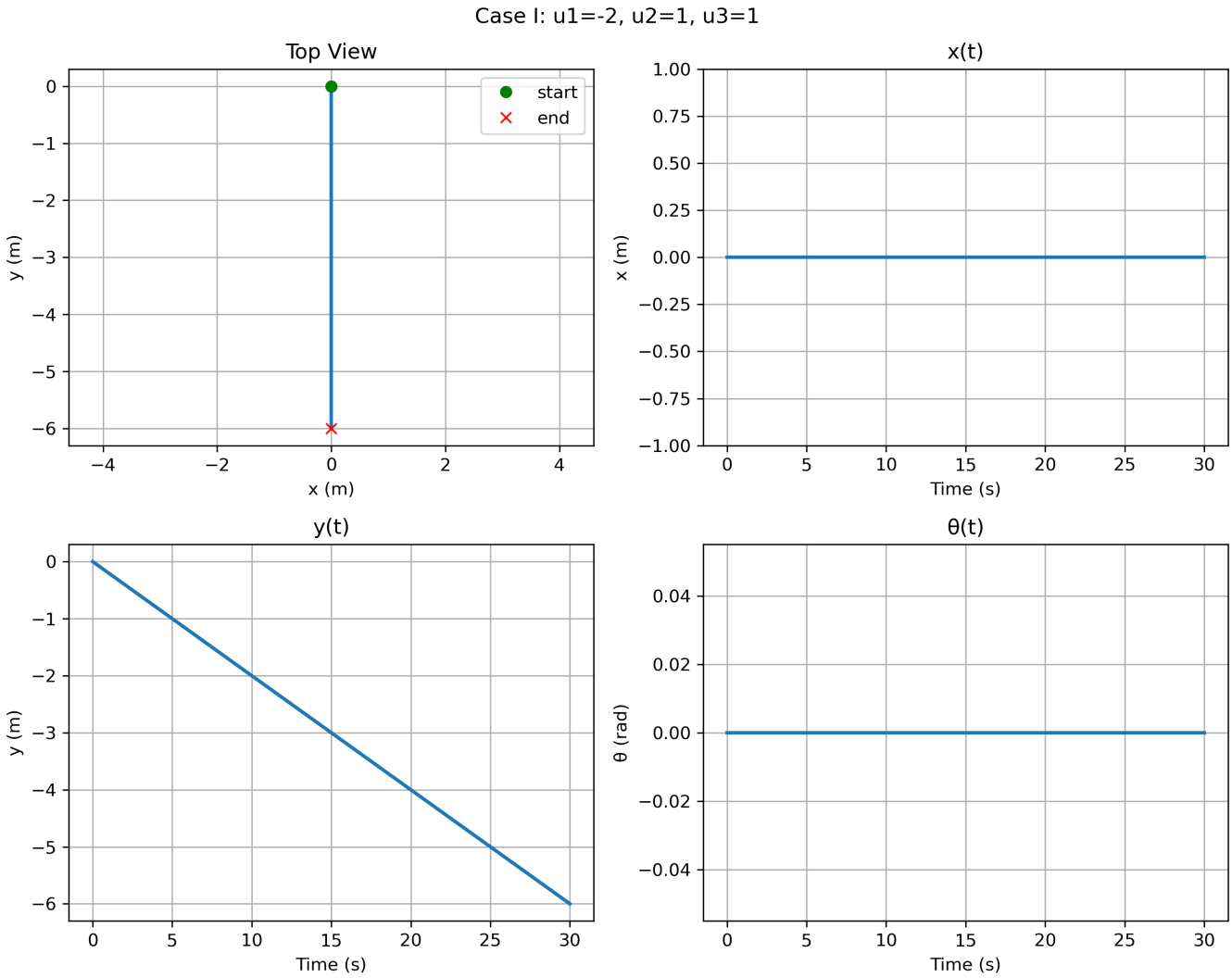


Figure 5: Simulation results for $u_1 = -2,\ u_2 = 1,\ u_3 = 1$.

Solution to (b) II.1

To generate a straight-line trajectory with a slope of 60°, the translational velocities must satisfy

$$\frac{\dot{y}}{\dot{x}} = \tan(60^\circ) = \sqrt{3}$$

Choosing a convenient velocity vector gives

$$\dot{q} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} 1 \\ \sqrt{3} \\ 0 \end{bmatrix}$$

where $\dot{\theta} = 0$ ensures that the robot maintains a constant orientation while moving.
Using

$$u = G(\theta)\dot{q}$$

and evaluating at $\theta = 0$,

$$u = \frac{1}{r} \begin{bmatrix} 0 & 1 & l \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} & l \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} & l \end{bmatrix} \begin{bmatrix} 1 \\ \sqrt{3} \\ 0 \end{bmatrix}$$

$$u = \frac{1}{r} \begin{bmatrix} \sqrt{3} \\ -\sqrt{3} \\ 0 \end{bmatrix}$$

Multiplying all wheel speeds by the same constant only changes the speed of travel and not the direction of motion. Therefore, a simpler equivalent control input is

$$u = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \text{ rad/s}$$

which produces the same straight-line trajectory with a slope of 60°.

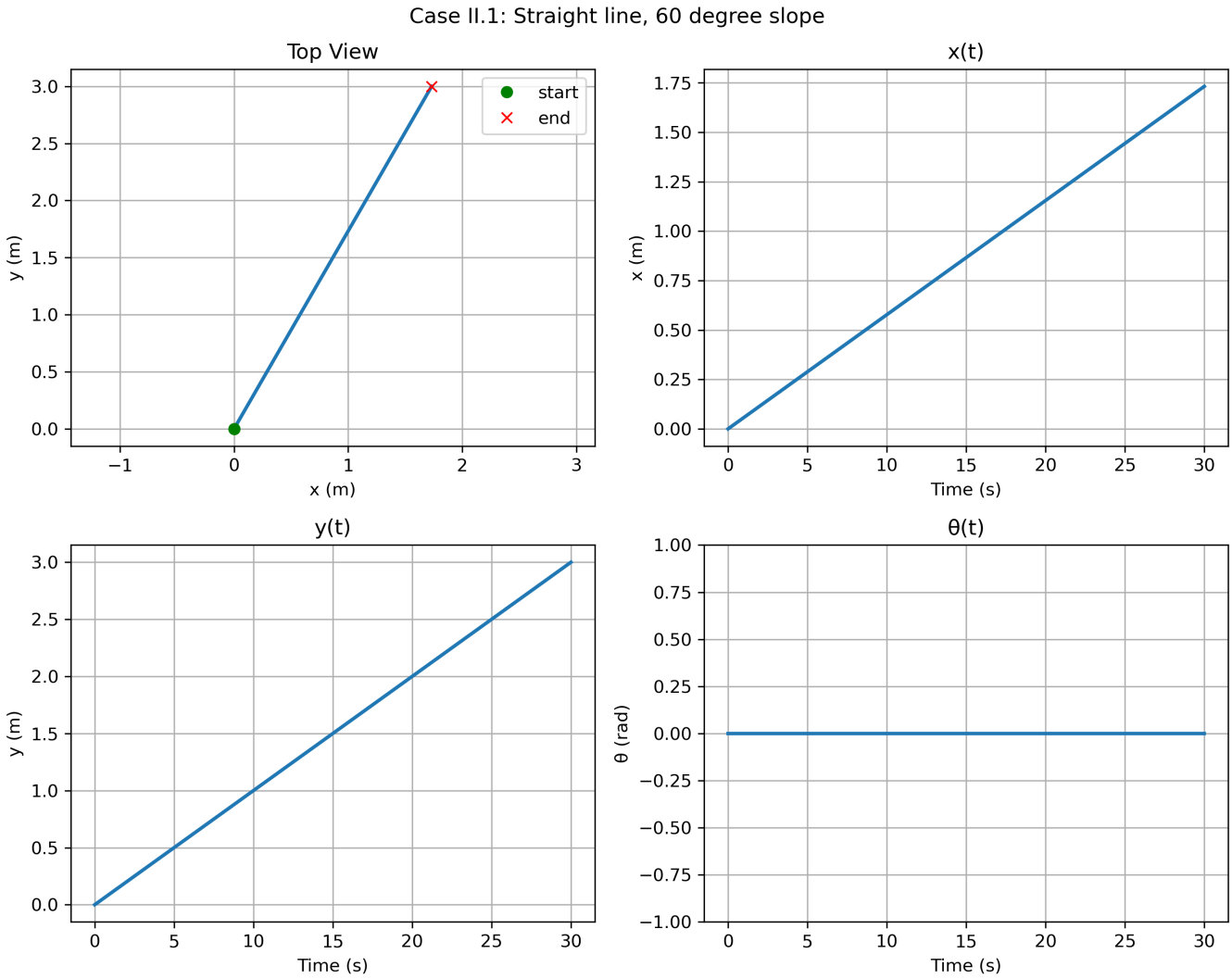


Figure 6: Simulation results for the 60° straight-line trajectory.

Solution to (b) II.2

For a circular trajectory with radius $R = 1\,\text{m}$, the linear and angular velocities are related by

$$v = R\dot{\theta}$$

Choosing

$$v = \dot{\theta} = 1$$

gives

$$\dot{q} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos(\theta) \\ \sin(\theta) \\ 1 \end{bmatrix}$$

where the translational velocity has magnitude

$$\sqrt{\dot{x}^2 + \dot{y}^2} = \sqrt{\cos^2(\theta) + \sin^2(\theta)} = 1\,\text{m/s}.$$

Using the relationship

$$u = G(\theta)\dot{q}$$

with $r = 0.1\,\text{m}$ and $l = 0.25\,\text{m}$, the required wheel speeds are

$$u = \frac{1}{0.1}\begin{bmatrix} \cos\left(\theta + \frac{\pi}{2}\right) & \sin\left(\theta + \frac{\pi}{2}\right) & 0.25 \\ \cos\left(\theta + \frac{7\pi}{6}\right) & \sin\left(\theta + \frac{7\pi}{6}\right) & 0.25 \\ \cos\left(\theta + \frac{11\pi}{6}\right) & \sin\left(\theta + \frac{11\pi}{6}\right) & 0.25 \end{bmatrix}\begin{bmatrix} \cos(\theta) \\ \sin(\theta) \\ 1 \end{bmatrix}.$$

Simplifying the resulting expressions gives

$$u_1 = \frac{5}{2},$$

$$u_2 = \frac{5}{2} - 5\sqrt{3},$$

and

$$u_3 = \frac{5}{2} + 5\sqrt{3}.$$

Therefore, the required wheel-speed control input is

$$u = \begin{bmatrix} \frac{5}{2} \\ \frac{5}{2} - 5\sqrt{3} \\ \frac{5}{2} + 5\sqrt{3} \end{bmatrix}\ \text{rad/s}.$$

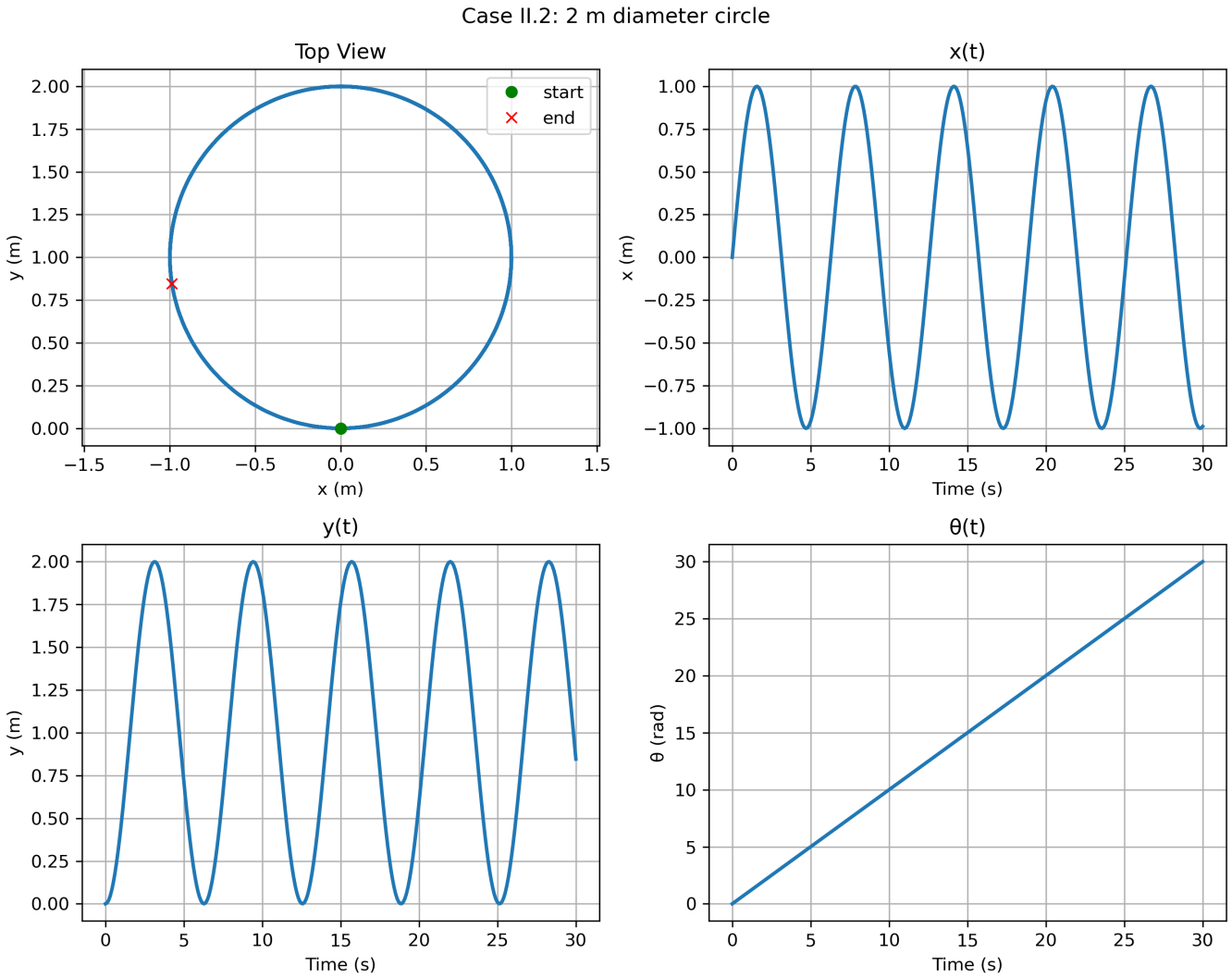


Figure 7: Simulation results for the 2 m diameter circular trajectory.